Numerical Analysis

Lab 7

Giuseppe Caliò

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Iterative methods for linear systems

This lab deals with the numerical solution of a linear system

$$\mathbb{A}\boldsymbol{x} = \boldsymbol{b}, \quad \text{for } \mathbb{A} \in \mathbb{R}^{n\times n}, \boldsymbol{b} \in \mathbb{R}^n$$

We will focus on dynamic iterative methods.

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Gradient method I

Solving a linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ with $\mathbb{A}$ **symmetric positive definite** is equivalent to solving the **minimization problem** of the quadratic form

$$\Phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathbb{A}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}.$$

The idea of the **preconditioned gradient method** is to compute the solution $\boldsymbol{x}^*$ with the sequence

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha^{(k)}\mathbb{P}^{-1}\boldsymbol{d}^{(k)}.$$

taking $\boldsymbol{d}^{(k)}$ to be the **steepest descent** direction for $\Phi(\boldsymbol{x})$ at $\boldsymbol{x}^{(k)}$ and a step length $\alpha^{(k)}$ such that $\Phi(\boldsymbol{x}^{(k+1)})$ is minimized. Hence these parameters are set **dynamically** to

$$\boldsymbol{d}^{(k)} = \boldsymbol{r}^{(k)} = \boldsymbol{b} - \mathbb{A}\boldsymbol{x}^{(k)}, \quad \mathbb{P}\boldsymbol{z}^{(k)} = \boldsymbol{r}^{(k)} \quad \text{and} \quad \alpha^{(k)} = \frac{{\boldsymbol{z}^{(k)}}^T\boldsymbol{r}^{(k)}}{{\boldsymbol{z}^{(k)}}^T\mathbb{A}\,\boldsymbol{z}^{(k)}}.$$

Choice of α in the dynamic Richardson method: optimality

Theorem 7.1 (Preconditioned Gradient (PG) method)

Let $\mathbb{A}$ and $\mathbb{P}$ be symmetric positive definite matrices. If

$$\alpha^{(k)} = \frac{\boldsymbol{z}^{(k)^T} \boldsymbol{r}^{(k)}}{\boldsymbol{z}^{(k)^T} \mathbb{A}\, \boldsymbol{z}^{(k)}}, \quad k \geq 0,$$

with $\boldsymbol{z}^{(k)} = \mathbb{P}^{-1}\boldsymbol{r}^{(k)}$ preconditioned residual, then the dynamic Richardson method $\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \mathbb{P}^{-1}\boldsymbol{r}^{(k)}$, is convergent for any initial guess and the convergence rate is the greatest possible.

Remark: the residual $\boldsymbol{r}^{(k)}$ is the **steepest descent** direction for $\Phi(\boldsymbol{x})$ at $\boldsymbol{x}^{(k)}$ and the acceleration parameters α_k are such that $\Phi(\boldsymbol{x}^{(k+1)})$ is minimized, with $\Phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathbb{A}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}$ (minimizing $\Phi(\boldsymbol{x})$ is equivalent to solve $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$.)

Gradient method I

Preconditioned Gradient method

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \mathbb{P}^{-1} \boldsymbol{r}^{(k)}$$

$$\begin{aligned}
\mathbb{P}\boldsymbol{z}^{(k)} &= \boldsymbol{r}^{(k)}, \\
\alpha_k &= \frac{{\boldsymbol{z}^{(k)}}^T \boldsymbol{r}^{(k)}}{{\boldsymbol{z}^{(k)}}^T \mathbb{A}\, \boldsymbol{z}^{(k)}} \\
\boldsymbol{x}^{(k+1)} &= \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{z}^{(k)} \\
\boldsymbol{r}^{(k+1)} &= \boldsymbol{r}^{(k)} - \alpha_k \mathbb{A} \boldsymbol{z}^{(k)} (\boldsymbol{b} - \mathbb{A}\boldsymbol{x}^{(k+1)})
\end{aligned}$$

Exercise 7.1

Consider the linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ with

$$\mathbb{A} = \begin{bmatrix} 2 & -1 & 0 & 1 \\ -1 & 4 & 2 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -1 & 1 & 5 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 6 \end{bmatrix}.$$

a. Can you apply the gradient method to the system?
b. Starting from the function for the Richardson method, implement the preconditioned gradient method in another function.
c. Apply the method with no preconditioner to the system in question, starting from $\boldsymbol{x}^{(0)} = \mathbf{0}$ and with a tolerance of 10^{-9}.

Exercise 7.2

Consider the following linear systems

$$\begin{cases} 2x - y = 1 \\ -x + 2y = 1 \end{cases} \qquad \begin{cases} x - y = 0 \\ -x + 20y = 19 \end{cases} \qquad \begin{cases} 3x + 0y = 3 \\ 0x + 3y = 3. \end{cases}$$

a. Verify that the coefficient matrices of the systems are symmetric positive definite.

b. For each system graphically visualize the iterative steps of the gradient method with respect to the contour lines of the quadratic form $\Phi(\boldsymbol{x})$ associated to the problem

$$\Phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T \mathbb{A}\boldsymbol{x} - \boldsymbol{x}^T \boldsymbol{b},$$

where $\mathbb{A}$ is the coefficient matrix and $\boldsymbol{b}$ the right-hand side of the system. What can you observe?

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Preconditioned Conjugate Gradient method

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \widetilde{\alpha}_k \mathbb{P}^{-1}\boldsymbol{p}^{(k)}, \quad \boldsymbol{p}^{(k)} \ s.t \ (\mathbb{A}\boldsymbol{p}^{(j)})^T \boldsymbol{p}^{(k)} = 0, i = 0 : k$$

$$\begin{aligned}
\widetilde{\alpha_k} &= \frac{{\boldsymbol{p}^{(k)}}^T \boldsymbol{r}^{(k)}}{{\boldsymbol{p}^{(k)}}^T \mathbb{A}\, \boldsymbol{p}^{(k)}}, \\
\boldsymbol{x}^{(k+1)} &= \boldsymbol{x}^{(k)} + \widetilde{\alpha}_k \boldsymbol{p}^{(k)}, \\
\boldsymbol{r}^{(k+1)} &= \boldsymbol{r}^{(k)} - \widetilde{\alpha}_k \mathbb{A}\boldsymbol{p}^{(k)} (\boldsymbol{b} - \mathbb{A}\boldsymbol{x}^{(k+1)}), \\
\mathbb{P}\boldsymbol{z}^{(k+1)} &= \boldsymbol{r}^{(k+1)}, \\
\beta_k &= \frac{\left(\mathbb{A}\boldsymbol{p}^{(k)}\right)^T \boldsymbol{z}^{(k+1)}}{\left(\mathbb{A}\boldsymbol{p}^{(k)}\right)^T \boldsymbol{p}^{(k)}}, \\
\boldsymbol{p}^{(k+1)} &= \boldsymbol{r}^{(k+1)} - \widetilde{\beta}_k \boldsymbol{p}^{(k)}
\end{aligned}$$

Exercise 7.3

Consider the linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$, with

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -8 \end{bmatrix} \quad \text{e} \quad \mathbf{x} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

- represent the level curves of the quadratic form linked to the gradient method
  Hint:to visualize the level curves of a function $f : \mathbb{R}^2 \to \mathbb{R}$,e.g. $f(x, y) = x^2 - y^2$, you can use the contour function: `contour(X,Y,f(X,Y))`
- when setting $\mathbf{x}^{(0)} = [-2, 2]^T$, how many iterations are required to reach the tolerance $toll = 10^{-6}$ by employing the gradient method? Represent the successive approximations $\mathbf{x}^{(k)}$ together with the level curves of $\Phi(\mathbf{z})$
- repeat the operations of the previous point, using the preconditioned gradient method with P= diag(A);
- repeat the operations of the previous point, using the conjugate gradient method (without preconditioning) and the preconditioned version (pcg MATLAB function).
- compare all the results

Newton method for non-linear systems I

Scalar case: $f(x) = 0$, f scalar function

$$x^{(k+1)} = x^{(k)} + \delta^{(k)}, \qquad \delta^{(k)} = -\frac{f(x^{(k)})}{f'(x^{(k)})},$$

where the correction $\delta^{(k)}$ is formally obtained by solving the linear equation

$$f'(x^{(k)})\, z = -f(x^{(k)})$$

Vectorial case: $\mathbf{f}(\mathbf{x}) = \mathbf{0}$, f vectorial function

$$x^{(k+1)} = x^{(k)} + \delta^{(k)},$$

where the correction $\delta^{(k)}$ is found by solving the linear system

$$\mathrm{J}\mathbf{f}(x^{(k)})\, z = -\mathbf{f}(x^{(k)}), \qquad \mathrm{J}\mathbf{f} \text{ Jacobian matrix of } \mathbf{f}$$

Newton method for non-linear systems II

Exercise 7.4

Consider the following system of equations

$$\begin{cases} x_1^2 + x_2^2 - 1 = 0, \\ sin(\pi x_1/2) + x_2^3 = 0. \end{cases}$$

Use the Newton method to find the solution of the system with a tolerance equal to 10^{-10}, by considering as initial guess $x_0 = -1$ and $y_0 = -1$ and $Nmax = 200$. [Hint: use the newtonsys.m function]

Newton method for non-linear systems III

Exercise 7.5 (Exam Sep 6, 2024)

Consider the non-linear system below, where $\mathbf{x} = [x_1, x_2]^T$ is the unknown vector:

$$\begin{cases} \dfrac{2x_1^2}{x_1^2+1} & = 1 \\ \dfrac{2x_1^2+2x_2^2}{x_1^2+x_2^2+1} & = 1 \end{cases}$$

Newton method for non-linear systems IV

Exam Sep 6, 2024

1. Write the nonlinear system in the form $\mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x})]^T = \mathbf{0}$. Represent $\mathbf{f}$ usign a Matlab® *anonymous function* that, upon receiving a column vector of 2 elements as input, returns a column vector of 2 elements containing the evaluation of $\mathbf{f}(\mathbf{x})$. Similarly, compute the Jacobian matrix $J(\mathbf{x})$ of $\mathbf{f}(\mathbf{x})$ and create the *anonymous function* that returns this matrix. Provide the MATLAB® instructions.
2. Use the provided `newtonsys.m` function to calculate the solution of the proposed nonlinear system. Use a tolerance of `tol=1e-6`, a maximum number of 1000 iterations, and the initial vector $\mathbf{x}^{(0)} = [3/2, 1/2]^T$. Then, repeat the calculation using the initial vector $\mathbf{x}^{(0)} = [-3/2, 1/2]^T$. Provide the obtained results and the number of iterations performed. Include the MATLAB® instructions used.
3. Comment on the results obtained in the previous step in light of the theory.
4. Let $e^{(k)} = \|\mathbf{x}^{(k)} - \alpha\|$ be the error associated with the k^{th} iteration, where α represent the root $(-1, 0)$. Using again the initial guess $\mathbf{x}^{(0)} = [-3/2, 1/2]^T$, construct a vector containing the errors $e^{(k)}$ obtained at each iteration. Report the MATLAB® code, as well as the values of $e^{(k)}$ for the first 4 iterations $k = 0, 1, 2, 3$ (including the initial guess). Report at least the first 3 significant figures.
5. Using the same tolerance and maximum number of iterations as above, use the `newtonsys.m` function again with $\mathbf{x}^{(0)} = [3, 1/2]^T$. Provide the obtained results and discuss them in light of the theory.

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Homework 7.1

Consider the following linear system $\mathbb{A}_n \boldsymbol{x} = \boldsymbol{b}_n$ with $n = 10, 15$, where matrix $\mathbb{A}_n$ and vector $\boldsymbol{b}_n$ are constructed with the MATLAB statements

$$\mathbb{A}_n = \texttt{eye(n) + triu(pascal(n),1)}, \quad \boldsymbol{b}_n = \texttt{[1:n]'}.$$

Consider now the following family of iterative methods with parameter α

$$\boldsymbol{x}^{(k+1)} = (\mathbb{I} + \alpha \mathbb{A}_n)\, \boldsymbol{x}^{(k)} - \alpha \boldsymbol{b}_n$$

where $\mathbb{I}$ is the identity matrix of order n.

a. Use MATLAB to determine if the methods obtained by setting $\alpha = 1, -1/2$ and -1 are convergent to the exact solution $\boldsymbol{x}^*$ of the system, in both cases $n = 10, 15$.
b. Implement the method for the cases in which convergence is assured. Find the solution $\boldsymbol{x}^*$ with tolerances 10^{-6} and 10^{-9} for both $n = 10, 15$ and indicate the number of performed iterations. Report the solutions in MATLAB `short e` format.
c. What about the case of $\alpha = -1$? Can you motivate the result of the previous point?
d. Was it possible to know the behaviour of the methods just looking at the structure of the iteration matrices $(\mathbb{I} + \alpha \mathbb{A}_n)$? How?

Homework II

Homework 7.2

Consider the linear system $\mathbb{A}_n \boldsymbol{x} = \boldsymbol{b}_n$, where $\mathbb{A}_n$ and $\boldsymbol{b}_n$ are obtained with the MATLAB commands

```
>> A = gallery('poisson',n);  b = A*ones(size(A(:,1)));
```

with $n = 3, 5, 10, 20$.

a. Write, in the same script as the rest of the exercise (as you are going to do the day of the exam), the gradient method with a convergence check on the normalized residual and returning the history of the norm of this latter. Solve the systems with $\mathbb{P} = \mathbb{I}$, a tolerance of 10^{-6} and $\boldsymbol{x}^{(0)} = \mathbf{0}$.
b. Represent the evolution of the residual norm vs the number of iterations. Represent also the number of iterations needed to reach the tolerance as a function of n.
c. Do you observe any relationship between the $\mathcal{K}(\mathbb{A}_n)$ and the convergence rate of the method? Justify the answer.

Homework 7.3

Consider the matrix obtained with:

```
>> B = rand(5) + diag(10*ones(5,1));
>> A = B*B';
```

a. Compute the solution of the system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ using the non preconditioned gradient method with right hand side $\boldsymbol{b} = \mathbf{1}$, initial guess $\boldsymbol{x}^{(0)} = \mathbf{0}$ and $\texttt{tol} = 10^{-i}$ for $i = 3, 6, 9, 12$ and 15.
b. Write, in the same script as the rest of the exercise (as you are going to do the day of the exam), the gradient method substituting the line in which the residual is updated with the instruction

   ```
   r = b - A*x;
   ```

   Solve the system using the original and the modified version of the function with a very small value for the tolerance, such as $\texttt{tol} = 10^{-20}$. Explain the obtained results.

Homework IV

Homework 7.4

Find the minimum of the quadratic form

$$\Phi(x, y) = x^2 - 4x + y^2 - y - xy.$$

References I